

Are 4mm super short dental implants a suitable adjunct in the rehabilitation of resorbed alveolar bone?

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REVIEW TITLE AND BASIC DETAILS

Review title

Are 4mm super short dental implants a suitable adjunct in the rehabilitation of resorbed alveolar bone?

Review objectives

The aim is to find out if super short 4mm long dental implants can be used to successfully support multi-implant borne prostheses with or instead of conventional implants.

the research questions are: are there more implant failures - marginal bone loss, peri implantitis, implant fractures - when using super short implants? and are there more prosthetic failures - pink white aesthetics, fractures?

Keywords

Conventional implants, Dental implants, Implant fractures, Marginal bone loss, Multi-implant borne prostheses, Peri-implantitis, Prosthetic failures, Resorbed alveolar bone, Super short dental implants

SEARCHING AND SCREENING

Searches

A literature review of randomised controlled trials will be conducted in PubMed (<http://www.ncbi.nlm.nih.gov/PubMed>), the Cochrane Library

(<http://www.cochranelibrary.com>) and ScienceDirect (<https://www.ScienceDirect.com>).

The search will take place on the 20th of February 2023. There are no other search restrictions.

Additional search strategy information can be found in the attached PDF document (link provided below).

Study design

Inclusion criteria: RCTs only, any language.

Exclusion criteria: non-RCTs.

ELIGIBILITY CRITERIA

Condition or domain being studied

Edentulous or partially edentulous resorbed mandible.

Population

Inclusion criteria: 18+ years old.

Exclusion criteria: autoimmune conditions, cancer, smoking 10+ cigarettes a day, space for only 1 implant.

Intervention(s) or exposure(s)

Ultra short dental implants - < 5mm in length.

Comparator(s) or control(s)

Conventional or short implants (5mm ≤ length).

Context

No other inclusion or exclusion criteria, implants are designed to be universal prosthetic devices for replacing the function of a missing tooth in a stable dentition. Dentition can also be stabilised by occlusal adjustment using implants although this is not advised with single implants, and two or more implants have to be splinted for this effect. An unstable dentition is when the jawbones are changing shape due to puberty, or bone turnover is affected by disease states. Therefore, age and medical conditions and smoking are the only patient related selection criteria, as well as the need for more than 1 implant next to each other.

OUTCOMES TO BE ANALYSED

Main outcomes

Implant and prosthesis survival and success criteria as measures.

Measures of effect

Measured as a relative risk.

Additional outcomes

Marginal bone levels in mm loss.

Measures of effect

Mean difference.

DATA COLLECTION PROCESS

Data extraction (selection and coding)

Study selection: 2 reviewers independently assessing studies for selection criteria with disagreements resolved by an arbitrator. decisions recorded in a Word document.

Data extraction: 2 reviewers one extracting the data and the other checking it over, with disagreements resolved by external arbitrator. Data for MBL, implant survival and implant success criteria will be collated. Due to differences in measures for implant success and survival the values might need to be weighted according to specific consensus by reviewers. missing data will be expertly assessed as to whether it is missing at random or not. if it is thought to be missing at random and less than 10% of the values are missing, regression imputation will be used. for more than 10% missing multiple imputation will be used. if possible bootstrapping techniques will be used. Study investigators will be contacted for missing data first. data will be recorded in an Excel spreadsheet. R will be used for management and statistical assessment.

Risk of bias (quality) assessment

Involvement with implant manufacturers, convolution of assessor/organiser roles. CASP and Jadad tools will be used.

PLANNED DATA SYNTHESIS

Strategy for data synthesis

For the meta-analysis of RCT data for marginal bone loss as well as implant success and survival measures for 4mm implants versus all other implants in a population of 18-year-olds, we plan to use a random-effects meta-analysis model to combine the results from individual studies. The random-effects model assumes that the true treatment effect may vary across studies, and incorporates both within-study and between-study variability in the estimation of the overall treatment effect.

We will explore statistical heterogeneity using the I^2 statistic, which estimates the proportion of total variation in study estimates that is due to heterogeneity rather than chance.

We plan to use the R software package for conducting the meta-analysis, specifically the metafor package, which provides a comprehensive set of functions for fitting and interpreting random-effects meta-analysis models. We will use forest plots to visualize the individual study estimates and the overall treatment effect, and we will also conduct sensitivity analyses to assess the robustness of the results to various assumptions and methodological choices.

For the analysis of marginal bone loss, we will extract mean differences and standard deviations from each study and calculate the pooled estimate using a random-effects meta-analysis model. For the analysis of implant success and survival measures, we will extract the number of events and total sample size from each study and calculate the pooled relative risk using a random-effects meta-analysis model.

Analysis of subgroups or subsets

We will also conduct subgroup analyses based on study characteristics such as sample size,

study quality, and follow-up time to explore potential sources of heterogeneity.

REVIEW AFFILIATION, FUNDING AND PEER REVIEW

Review team members

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Funding source

Self funded

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TIMELINE OF THE REVIEW

Review timeline

Start date: 20 February 2023. End date: 25 March 2023

Date of first submission to PROSPERO

19 February 2023

Date of registration in PROSPERO

22 February 2023

CURRENT REVIEW STAGE

Publication of review results

The intention is to publish the review once completed. The review will be published in English

Stage of the review at this submission

Review stage	Started	Completed
Pilot work		
Formal searching/study identification		
Screening search results against inclusion criteria		
Data extraction or receipt of IP		
Risk of bias/quality assessment		
Data synthesis		

Review status

The review is currently planned or ongoing.

ADDITIONAL INFORMATION

PROSPERO version history

- Version 1.0 published on 22 Feb 2023

Review conflict of interest

None known

Country

Poland, Spain

Medical Subject Headings

Alveolar Bone Loss; Bone Diseases, Metabolic; Bone and Bones; Esthetics; Humans; Peri-Implantitis

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